

PTB-XL ECG Classification: A Comparative Study of ML and DL Techniques

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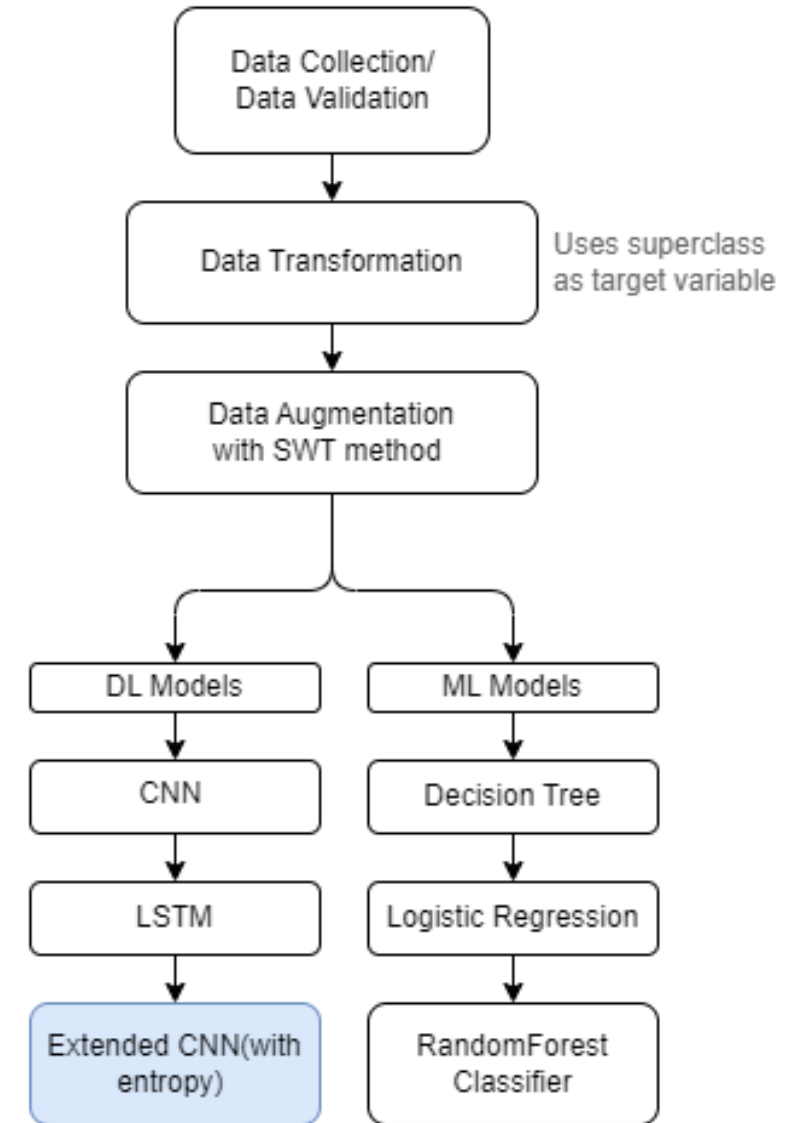
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1. Introduction

- **OBJECTIVE:** To compare the ML and DL models on ECG data.
 - Prepare the optimal model for accurate ECG classification.
- **WHY? :**
 - There 4 main NCDs (Non-Communicable Diseases).
 - **Cardiovascular Disease**, Diabetes, Chronic Respiratory Disease, Cancer.
 - Total Deaths by NCDs **41 million** annually
 - Deaths by CVD **77%** or **17.9 million**
- **Scope & Dataset:**
 - Comparison of ML and DL for identifying the effective solution.
 - Dataset, using the PTB-XL standard 12-lead ECG recordings.

Work Done Flow Chart



2. Problem Statement

Cardiovascular diseases (CVDs) are the leading cause of death globally, necessitating accurate and timely diagnosis to save lives. Electrocardiograms (ECGs) are essential tools in diagnosing CVDs.

Challenges:

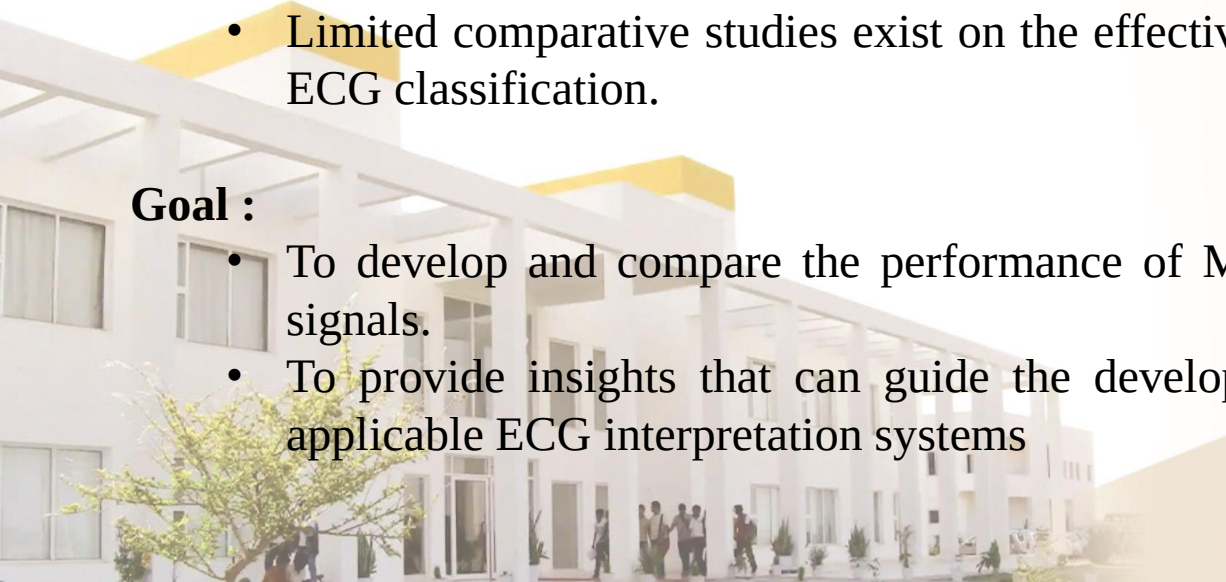
- The traditional way of interpreting ECGs depends a lot on human expertise, which can be subjective and prone to errors.
- Manual interpretation is time-consuming, so an automated and reliable system can help health professionals.

Need/Gap:

- Limited comparative studies exist on the effectiveness of various ML and DL models for ECG classification.

Goal :

- To develop and compare the performance of ML and DL models for classifying ECG signals.
- To provide insights that can guide the development of more advanced and clinically applicable ECG interpretation systems



3. Methodology

- **Data Preprocessing:**

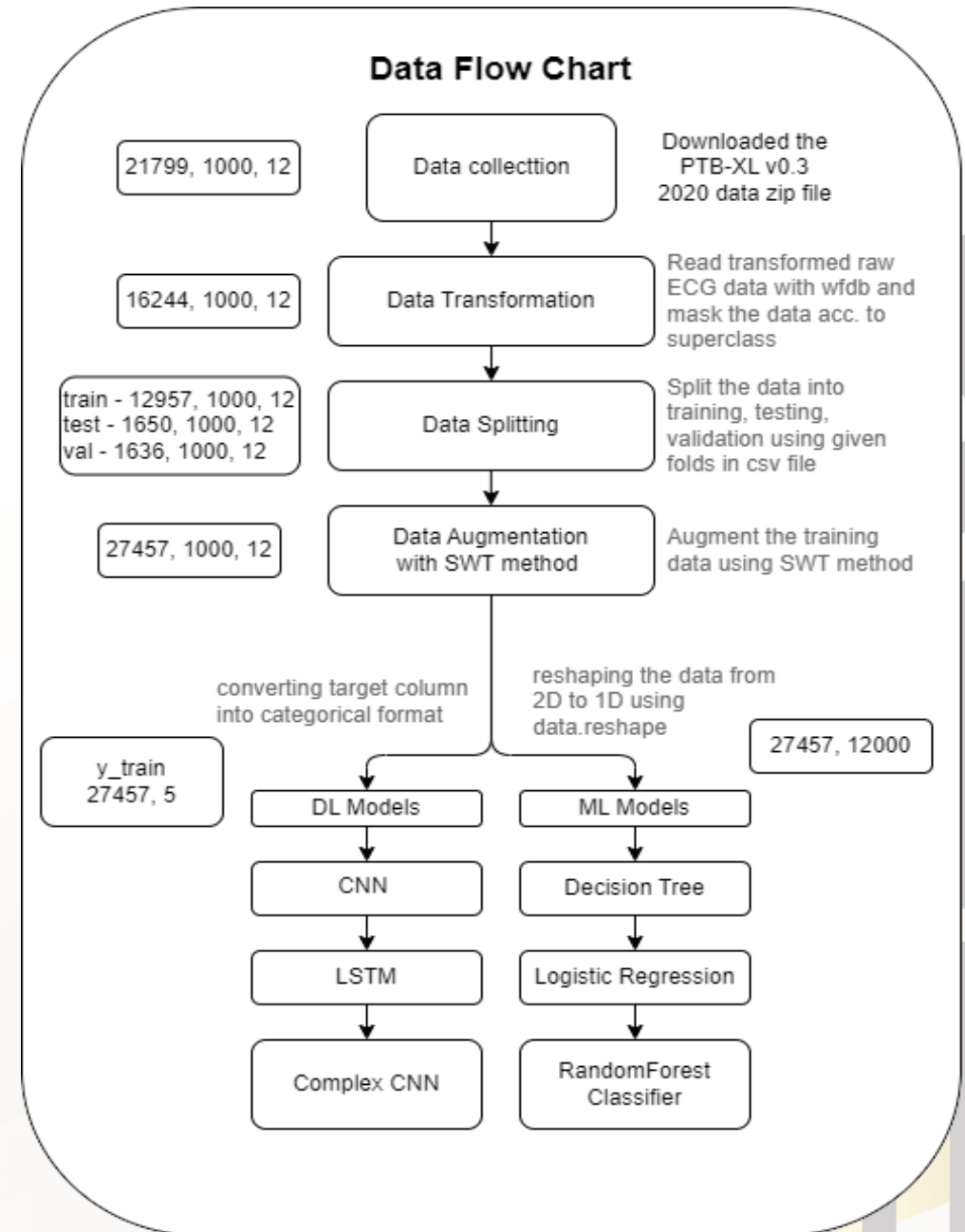
- Simplified the original 24 diagnostic super classes into 5 major super classes for easier classification.
- Uses data augmentation techniques to boost training sample diversity and to handle the class imbalance.

- **Model Selection:**

- **ML Models:** Decision Tree, Random Forest, Logistic Regression.
- **DL Models:** Simple CNN, LSTM, Complex CNN(ECG-Classifier).

- **Training Process:**

- Split the dataset into training, testing, and validation sets using the predefined folds provided in the PTB-XL dataset. Split the dataset into training, testing, and validation sets using the predefined folds provided in the PTB-

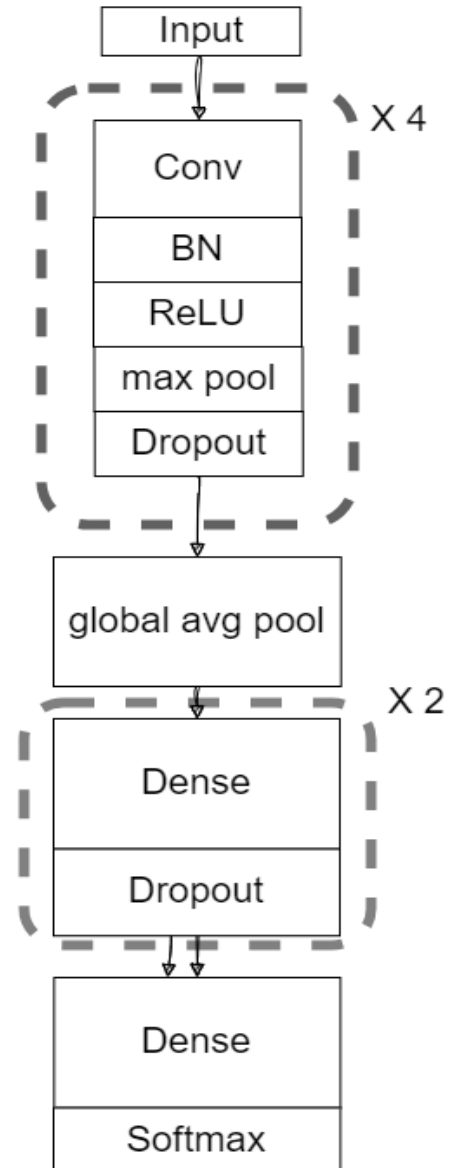


Data Flow Chart

4. Model Architecture

- **Input Shape:** (1000, 12) – 1000(10second x 100hz) time steps with 12 leads.
- **Convolutional Layers:**
 - Multiple 1D convolutional layers with filters increasing from 128 to 1024.
 - Kernel sizes: 15, 11, 7, and 5.
 - Each convolutional layer is followed by Batch Normalization and ReLU activation.
 - MaxPooling layers to reduce dimensionality.
- **Dropout Layers:**
 - Applied after each block of convolutional layers to prevent overfitting (Dropout rate: 0.4)
- **Global Average Pooling:**
 - Aggregates the features extracted by convolutional layers
- **Dense Layers:**
 - Two dense layers with 1024 and 512 units respectively, both with ReLU activation and dropout.
 - Final dense layer with softmax activation for classification into 5 classes

Complex CNN Model



- **Recall (Sensitivity):** Ratio of true positive predictions to the actual positives.
- **F1-Score:** Harmonic mean of precision and recall.
- **AUC-ROC:** Area under the receiver operating characteristic curve, measuring the ability to distinguish between classes.

Training process:

- Split the dataset into training, testing, and validation sets using the predefined folds provided in the PTB-XL dataset.

Evaluation Metrics:

- **Accuracy:** Proportion of correctly classified samples.
- **Precision:** Ratio of true positive predictions to the total predicted positives.
- **Recall (Sensitivity):** Ratio of true positive predictions to the actual positives.
- **F1-Score:** Harmonic mean of precision and recall.
- **AUC-ROC:** Area under the receiver operating characteristic curve, measuring the ability to distinguish between classes.

Confusion Matrix

Predicted	Actual					
	CD	HYP	MI	NORM	STTC	
CD	871 52.79%	95 5.76%	61 3.70%	60 3.64%	28 1.70%	1115
HYP	12 0.73%	140 8.48%	17 1.03%	8 0.48%	1 0.06%	178
MI	16 0.97%	16 0.97%	160 9.70%	7 0.42%	2 0.12%	201
NORM	2 0.12%	3 0.18%	0 0.00%	106 6.42%	1 0.06%	112
STTC	11 0.67%	2 0.12%	4 0.24%	3 0.18%	24 1.45%	44
	912	256	242	184	56	1653

Confusion Matrix of ECG-Classifer Model

6. Result

Machine Learning Model:

- Decision Tree: 52% accuracy, 58% AUC-ROC.
- Logistic Regression: 40% accuracy, 50% AUC-ROC.
- **Random Forest: 54% accuracy, 62% AUC-ROC.**

Deep Learning Model:

- Simple CNN: 71% accuracy, 85% AUC-ROC.
- LSTM: 73% accuracy, 87% AUC-ROC.
- **Complex CNN(ECG-Classifer): 80% accuracy, 90% AUC-ROC.**

Table 3: Performance Comparison of Different Models for ECG Classification

Models	Accuracy(%)	Precision(%)	Recall(%)	F1-Score(%)	AUC-ROC(%)
CNN	71	73	66	69	85
LSTM	73	78	71	72	87
ECG-Classifier	80	80	76	78	90
Logistic Regression	40	21	22	21	50
Random Forest Tree	54	25	24	23	62
Decision Tree	52	24	24	22	58

7. Conclusion

- The Complex CNN model outperformed traditional ML models and other DL models in classifying ECG signals, with an accuracy of 80% and a ROC-AUC of 90%.
- This study highlights the potential of deep learning models, particularly CNNs, in automated ECG classification tasks
- Further research is needed to enhance the model's performance, possibly by incorporating more sophisticated architectures such as transformer models.
- Real-world clinical trials are essential to validate the model's applicability in diverse and practical healthcare environments
- The insights garnered from this comparative analysis can pave the way for advancements in ECG-based diagnosis and contribute to the effective utilization of machine learning in the healthcare industry.